MG MGHUB · DPD CONTROL TOWER

Platform load & cost map

Firebase, Supabase and Cloudflare — what each layer is billed for, the measured read and write numbers behind it, what a performance drop looks like on each one, and the order to fix them.

Firebase · Firestore + Functions | Supabase · paused | Cloudflare · Workers + DO + R2

Source docs/perf/platform-load-map.md · **Runtime** musallam-delivery-prod · **Generated** 2026-10-17

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Purpose: one document that keeps the hotspot diagram, states what each layer costs and why, records the *measured* numbers with their source file, and names what a performance drop on each layer looks like.

Read this first: Firestore bills **per document returned**, not per query. A `limit(2000)` query matching 500 rows costs 500 reads; six `where` clauses cost the same as two. `count()` / `sum()` bill **1 read per 1,000 index entries matched**. Writes bill at roughly **3–6× the read rate**. That single asymmetry is why this map separates "cheap call, high volume" from "expensive call, low volume" — they need opposite fixes.

Companions: `firestore-billing-explained.md` · `firestore-cost-baseline.md` · `firestore-read-reduction-plan.md` · `firestore-hotspots.md`

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0. The three layers in one line each

Layer	Role today	Billed by	Biggest line
Firebase (Firestore + Functions, `musallam-delivery-prod`)	Source of truth for every rider, delivery, shift, money row; compute for all 281 callables	documents read / written / stored, Cloud Run instance-seconds	document reads
Supabase (`eoksxkdssptgyqyywdju`)	**Paused.** No admin or Worker call reaches it; 0 imports in `src/`	nothing while paused	— (historically: Postgres DB time + egress)
Cloudflare (`dpd-live`, `dpd-images`, `dpd-esign-r2`)	Realtime edge rail, image resizing, private object storage	Worker + Durable Object requests, R2 operations, R2 storage	DO requests

Legend for every diagram below:

Colour	Meaning
🔴 red	primary contributor — fix this first
🟠 amber	second tier — contributes, bounded
🟢 green	controlled / no live traffic

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1. Firebase

1.1 Diagram

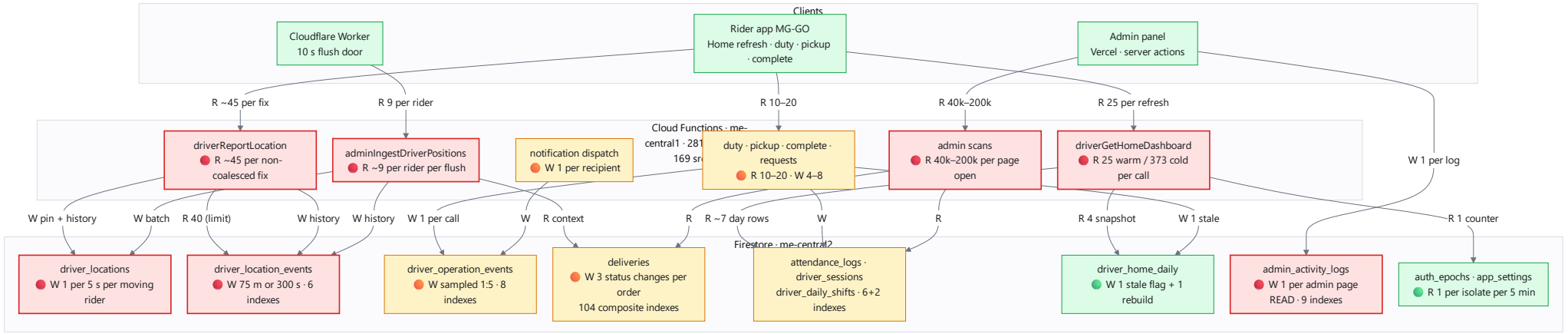

Clients
Cloudflare Worker
10 s flush door
Rider app MG-GO
Home refresh · duty · pickup
· complete
Admin panel
Vercel · server actions
R ~45 per fix
R 9 per rider
R 10–20
R 40k–200k
R 25 per refresh
driverReportLocation
R ~45 per non-
coalesced fix
adminIngestDriverPositions
R ~9 per rider per flush
notification dispatch
W 1 per recipient
Cloud Functions · me-
central1 · 281
169 sr
duty · pickup · complete ·
requests
R 10–20 · W 4–8
admin scans
R 40k–200k per page
open
driverGetHomeDashboard
R 25 warm / 373 cold
per call
W 1 per log
W pin + history
W batch
R 40 (limit)
W history
W history
W 1 per call
W
R context
R
R ~7 day rows
W
R
R 4 snapshot
W 1 stale
R 1 counter
Firestore · me-central2
driver_locations
W 1 per 5 s per moving
rider
driver_location_events
W 75 m or 300 s · 6
indexes
driver_operation_events
W sampled 1:5 · 8
indexes
deliveries
W 3 status changes per
order
104 composite indexes
attendance_logs ·
driver_sessions
driver_daily_shifts · 6+2
indexes
driver_home_daily
W 1 stale flag + 1
rebuild
admin_activity_logs
W 1 per admin page
READ · 9 indexes
auth_epochs · app_settings
R 1 per isolate per 5 min

FIGURE 1

1.2 Measured read cost — `driverGetHomeDashboard`

Live `perf:"handler"` line, 2026-10-17:
`{"name":"driverGetHomeDashboard","docsRead":25,"reads":17,"ms":1201}`

Part	Before	After	How measured
delivery rules + scopes + restaurants	371	**0** (isolate memory)	`count()`
incentive scopes / tiers / banners	0–688 by data	**0**	`count()`
rider session history (read twice)	82	**1** (`orderBy + limit(1)`)	block log
rider attendance day rows (read twice)	32	**~7**	block log
rider's own week of deliveries	75 (two scans)	**0 docs** (one `count()`)	block log
lifetime counts, compliance, pins, shifts	25	**25** (real data)	unchanged
documents per call	**528**	**25**	handler line
billed reads incl. aggregates	**528**	**~34**	25 + 9 aggregates

Reduction **95% on documents, ~94% on billed reads**. `reads` ≠ `docsRead`: `reads` is round trips (a latency signal), `docsRead` is the bill.

Cold is a different number. The 373 is the configuration fill, paid once per isolate per 5-minute window instead of once per call:

Calls per isolate per window	Average docs per call	Reduction
1	~373	29%
5	~95	82%
20	~43	92%
50	~32	94%

A working fleet sits at the bottom row. **Never judge this call from one hand-made probe** — take the p50 over a busy hour, or the second call.

Volume is the half the cache does not fix. At ~34 billed reads: `3,000 calls/day ≈ 102k reads/day`. Before the fix the same volume was ~1.6M/day. The remaining lever is *call frequency* — the snapshot plus an FCM `record_type: 'home'` refresh and one watcher on `drivers/{uid}` — not document count, which is already near the floor of what the rider's own data requires.

1.3 The GPS path — the largest un-cached read

`adminIngestDriverPositions` runs `ingestDriver()` per rider per flush (`fleet-ingest.ts` L280–290):

```
const [assigned, zone, history] = await Promise.all([
  loadAssignedRestaurantContext(args.driverId, args.driver),  // junction + restaurants + geofences
  loadDriverZone(zoneId),                                     // 1
  lastHistory(args.driverId),                                 // orderBy + limit(1)
]);
```

`loadAssignedRestaurantContext` (`driver-deliveries.ts` L356–396) reads the `driver_restaurants` junction, then `getAll` on the restaurants, then one `restaurant_geofences` query per 30-restaurant chunk. For a rider with two restaurants and three geofences that is **~2 + 2 + 3 + 1 + 1 = 9 documents**, and **nothing is memoised** — the same rows are re-read for the same rider on the next flush, ten seconds later.

`driverReportLocation` (the edge fallback) is worse per call and cheaper per document: `lastHistoryEvent` uses `limit(40)` and sorts in memory (`driver-location.ts` L160–178) — **40 reads to learn one row** — and both `deliveryExists` checks run *before* the coalesce test, so even a suppressed fix pays them. `fleet-ingest.ts` already has the correct shape at L193–200.

`POSTGRES_FLUSH_MS` 10000 (`wrangler.jsonc` L29) sets the cadence. **Model, not measured:** 900 riders × 8 h on duty = 2,880 flushes each → 2.6M flush-events/day → ~23M context reads/day if every flush carries a live point. Real traffic is lower because idle riders heartbeat less and the room batches, but the order of magnitude is what makes this the #2 line on the invoice.

1.4 Writes

Writer	Rate	Basis
`driver_locations`	1 write / 5 s / moving rider	`PIN_WRITE_MIN_MS` 1 s hard cap, Worker `FLUSH_MIN_GAP_MS` 5 s
`driver_location_events`	1 row / 75 m **or** 300 s	`HISTORY_METERS` / `HISTORY_SECONDS`
`driver_operation_events`	1 row per call, `location` sampled 1:5	`_shared.ts` L210–214, ~74 call sites
`admin_activity_logs`	**1 row per admin page read**	`logAdminRead`, 43 sites
`driver_home_daily`	1 stale flag + 1 rebuild per change	`driver-home.ts` L1776–1805
batch mutations	400 per batch	`BATCH_LIMIT`

Writes are dominated by the 1 Hz pipeline, which is a **product decision** — a map that glides needs fixes at 1 Hz. No read optimisation touches it. What *can* move is history cadence (`75 m / 300 s`) and retention (defaults now 30 / 90 / 7 days), both of which are configuration.

`admin_activity_logs` is the one write with no operational value at 1:1: reading a page is a write. Sampling read logs, or keeping only exports and decisions, removes a line that grows with *supervision*, not work.

1.5 Indexes — 186 composite, 104 on `deliveries`

Collection	Indexes	Note
`deliveries`	**104**	56% of the file; prefix/search + status/zone/partner/date filters
`admin_activity_logs`	9	append-only; read-side filters
`driver_operation_events`	8	sliding window queries
`fleet_events`	8	edge-derived events
`notification_dispatch_items`	7	per-recipient status
`driver_telemetry_events`	7	14-day retention
`attendance_logs`	6	date pages
`driver_location_events`	6	last-event + day route
`notification_campaigns`	4	list filters
`driver_wallet_entries`	3	status/date
`driver_sessions`	2	latest session
`driver_change_events`	2	per-driver history
20 other collections	20	1–1

Indexes do not change the billed document-write count — Firestore bills per document, not per index entry. They change **storage** and **write latency**: a delivery status change rewrites every index entry whose fields are in that document, and `deliveries` is written ~3× per order. At 900 riders this is felt on the verify queue, not on the invoice.

Two index facts learned here: an admin query with no matching composite degrades to a full scan (the 2026-10-02 `deliveries` entry: 114 MB re-read to return 50 rows), and an index whose sort direction does not match the emitted `ORDER BY … NULLS LAST` is never used. Both look like "the query is slow", not like "an index is missing".

1.6 Where the latency comes from

Cause	Evidence
sequential round trips, not payload	8 sequential fleet-wide reads → 1 counter read; block logs show `ms: 0`
function region ≠ database region	functions `me-central1`, Firestore `me-central2` — every hop crosses
cold isolate	373 docs + ~3.2 s first call vs 25 docs + ~1.2 s warm
admin scan volume	one Order Reconciliation open ≈ 50k–200k documents
missing/!=sort index	falls back to a bounded scan — slower, not broken

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2. Supabase — paused, and what that means for the bill

Status: project `eoksxkdssptgyqyywdju` is **paused**. `src/` contains **0** `@supabase` / `lib/supabase` imports. `@supabase/ssr` and `@supabase/supabase-js` remain in `package.json` with no consumer — a dependency to remove, not a service to bill.

2.1 Diagram

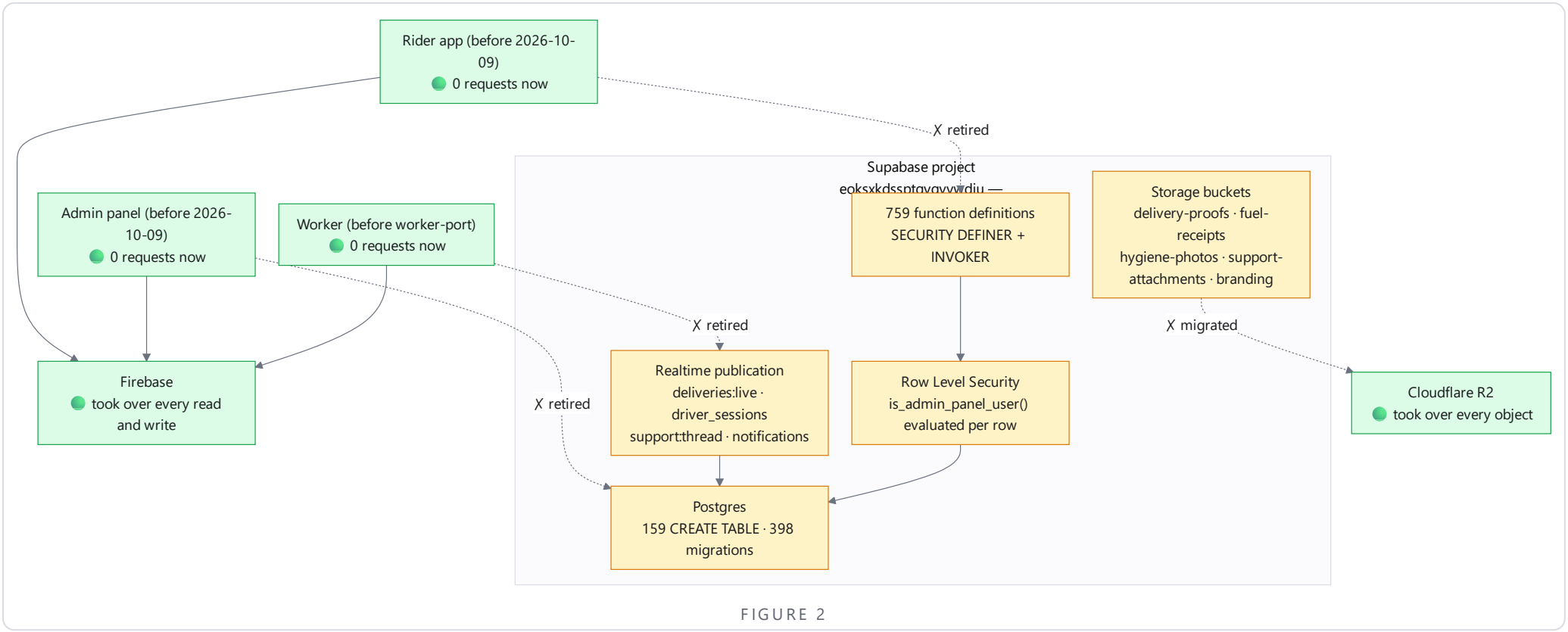

Rider app (before 2026-10-09)
0 requests now
X retired
Supabase project
eoksxkdssptqvgvvwdiu —
Admin panel (before 2026-10-09)
0 requests now
Worker (before worker-port)
0 requests now
759 function definitions
SECURITY DEFINER + INVOKER
Storage buckets
delivery-proofs · fuel-receipts
hygiene-photos · support-attachments · branding
X retired
X migrated
Firebase
took over every read and write
X retired
Realtime publication
deliveries:live · driver_sessions
support:thread · notifications
Row Level Security
is_admin_panel_user()
evaluated per row
Cloudflare R2
took over every object
Postgres
159 CREATE TABLE · 398 migrations

FIGURE 2

2.2 Relevant data

Fact	Number	Consequence today
Migration files kept in repo	**398**	history, not traffic
Tables ever created	**159**	schema knowledge only
Function definitions	**759**	RPC names still referenced in old notes
Live calls from `src/`	**0**	paused project costs nothing to run
Storage buckets still described in code	5	superseded by R2
Dead npm deps	2 (`@supabase/ssr` , `supabase-js`)	bundle surface, not cost

2.3 Why this layer stays in the document

It is the **counter-example that proves the rule**. Before the cutover this was the expensive tier, and every one of its problems was "the database did more work per request than the request needed":

- RLS policies called `is_admin_panel_user()` **per row**, so eight policies over a 184,282-row scan evaluated the check ~1.4 M times to return 50 rows.
- Server-side "list" RPCs ran the same statement 312k times at a 4.1 s mean — **1.28 B ms** for one statement, ~1.46 B ms for the deliveries family.
- `driver_sessions` had only a primary key: **4.08 M sequential scans / 77 B tuples read** to answer "is this rider online".
- Phones produced **~9 M requests/day at 25 % success** (2.7 M `app_settings` polls every 5 s, 2.1 M location writes, 1.25 M JWT validations).

The Firebase tier repeated exactly the same shape (Home re-reading 528 documents per call) and was fixed the same way: **serve from memory, aggregate instead of scan, bound the window**. If Supabase were ever resumed, the bill returns with per-row RLS, realtime publication and egress all intact — nothing in that diagram is a fallback, only history.

If reactivated, the cost drivers would be: compute (shared vs dedicated), database size, egress, and realtime concurrent connections. **If retired properly:** drop the two npm deps, keep the migration ledger as reference, and leave the project paused rather than deleted until the last Firebase-only quarter has closed.

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3. Cloudflare

Three Workers, one R2 bucket. This is the layer with the **clearest fixed ceiling**, because Workers bills per request and R2 bills per operation — no per-document amplification exists here.

3.1 Diagram

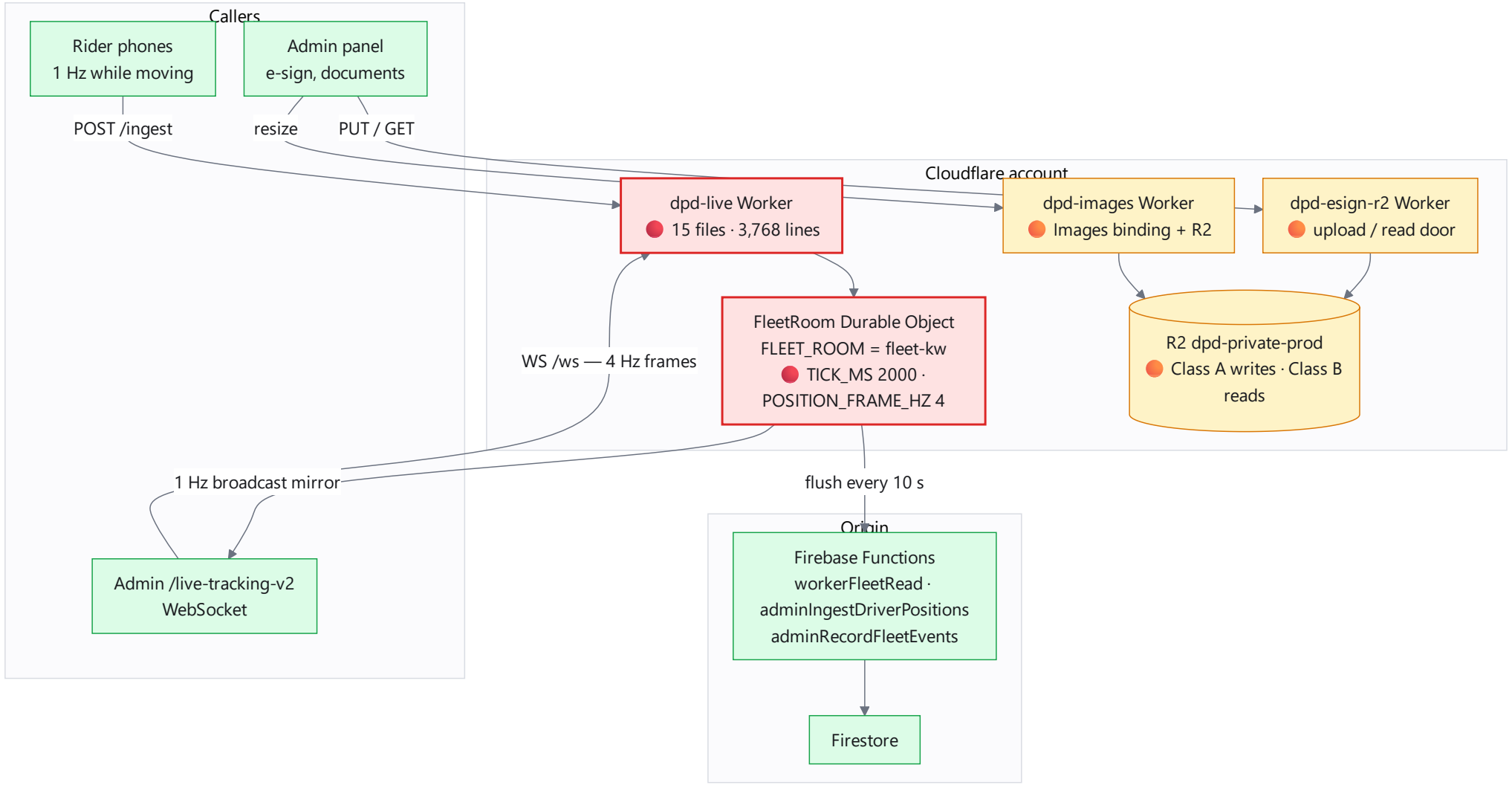

Callers
Rider phones
1 Hz while moving
Admin panel
e-sign, documents
POST /ingest
resize
PUT / GET
Cloudflare account
dpd-live Worker
15 files · 3,768 lines
dpd-images Worker
Images binding + R2
dpd-esign-r2 Worker
upload / read door
WS /ws — 4 Hz frames
FleetRoom Durable Object
FLEET_ROOM = fleet-kw
TICK_MS 2000 ·
POSITION_FRAME_HZ 4
R2 dpd-private-prod
Class A writes · Class B
reads
1 Hz broadcast mirror
flush every 10 s
Origin
Admin /live-tracking-v2
WebSocket
Firebase Functions
workerFleetRead ·
adminIngestDriverPositions
adminRecordFleetEvents
Firestore

FIGURE 3

3.2 Request accounting — measured, 16–17 Aug 2026

The free plan's ceiling is **100,000 requests/day, and it counts Worker *and* Durable Object requests together**. Measured from the GraphQL analytics API:

Component	Requests/day	Share
Durable Object (mostly the 2 s tick)	**96,692**	61%
Worker (one per ingest POST)	**60,924**	39%
Total	**157,616**	**158% of free tier**

The DO tick alone is `86,400 / 2 s = 43,200` requests/day — **43% of the entire free quota, burnt whether or not anyone is on duty**. The rest pairs ~1:1 because one ingest POST costs one Worker request plus the DO fetch it forwards.

The result is not hypothetical: on 17 Aug the Worker returned **1027** (daily request cap), the WebSocket closed **1006**, and the panel's connection pill fell to `mirror` — the three-rail fallback working as designed, but the map was no longer live. It is intermittent, not an all-day lockout: 1027 at 10:12 UTC, serving normally by 10:42.

Why trimming cannot fix it: a 5 s tick still lands at ~131k/day. The only shape that fits the free tier is moving ingest onto the established WebSocket (messages on an open socket cost **0** Worker requests and count **20:1** at the DO), which is a Worker + app change.

What it costs to stop caring: Workers Paid prices this fleet at about **$5.60/month** — ~3.7 M billable DO requests ≈ **$0.56** over the $5 minimum, with ~34,000 GB-s of duration inside the 400,000 included. That is the honest trade: one coffee against a map that is live on the free plan's terms.

3.3 R2 and the other two Workers

Item	Billed by	Notes
`dpd-private-prod`	Class A (writes) / Class B (reads), storage	eSign PDFs, signature PNGs, driver documents, request attachments
`dpd-esign-r2`	Worker requests	upload/read door; **403 on S3 falls through to this Worker**
`dpd-images`	Worker requests + Images	resizing bound to the same bucket
Egress	**$0**	R2's structural advantage over GCS/S3

The R2 layer is also where a *cost* becomes a *correctness* symptom: when S3 keys are absent or return 403, `esignComposeSignedDocument` and `adminApiHeavy` fall back to the eSign Worker. That fallback is why compose keeps working — and why a missing key looks like latency rather than an error.

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4. Combined — one GPS fix, one Home refresh, one page open

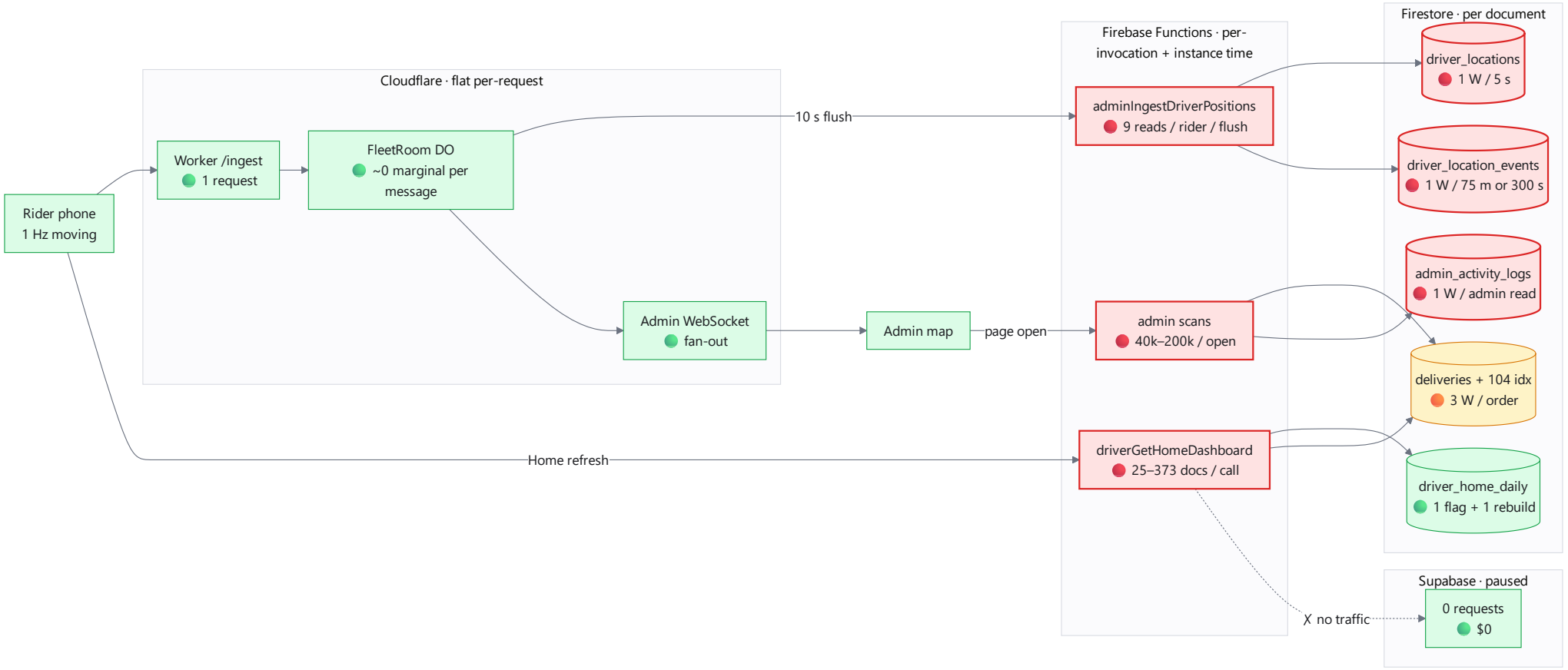

Cloudflare · flat per-request
Worker /ingest
1 request
FleetRoom DO
~0 marginal per message
Admin WebSocket
fan-out
Rider phone
1 Hz moving
10 s flush
Firebase Functions · per-invocation + instance time
adminIngestDriverPositions
9 reads / rider / flush
Admin map
page open
admin scans
40k–200k / open
driverGetHomeDashboard
25–373 docs / call
Home refresh
X no traffic
Firestore · per document
driver_locations
1 W / 5 s
driver_location_events
1 W / 75 m or 300 s
admin_activity_logs
1 W / admin read
deliveries + 104 idx
3 W / order
driver_home_daily
1 flag + 1 rebuild
Supabase · paused
0 requests
$0

FIGURE 4

4.1 Who pays for what

Cost line	Layer	Unit	Dominated by	Built-in ceiling?
Document reads	Firebase	per doc	Home volume, ingest context, admin scans	no — scales with calls × docs
Document writes	Firebase	per doc	1 Hz pin, history rows, audit logs	no
Storage + deletes	Firebase	per GiB / per doc	`driver_location_events`, `driver_operation_events`	yes — retention 30/90/7
Instance time	Firebase	Cloud Run vCPU·s + GiB·s	scan lane, cold fills	partly — `minInstances` 0
Egress	Firebase	per GiB	admin exports, proofs	no
Worker requests	Cloudflare	per request	DO tick, one per ingest	**yes — 100k/day free, then $5**
DO requests	Cloudflare	per request	2 s tick + ingest fan-in	**yes**
R2 ops / storage	Cloudflare	per 1k ops / GiB	eSign PDFs, document reads	yes — paid, linear
Postgres / egress / realtime	Supabase	paused	nothing	n/a

4.2 The three economics are not the same shape

- **Cloudflare is stepwise.** Free → 1027 at a fixed number, then a flat $5. It fails *visibly* (WS close, pill changes) and never runs away.
- **Firebase is linear per document.** It fails *silently*: the invoice moves, the app still works, and the only signal is `docsRead`. This is the layer that needs a measurement habit, not a spend cap.
- **Supabase was per-row-work.** Its failure mode was latency collapse at volume (1.28 B ms on one statement), which is why it was replaced rather than tuned.

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5. What a performance drop looks like, by layer

Symptom	Layer	Most likely cause	Fix
Home takes 3 s on first load, 1.2 s after	Firebase	cold isolate filling 373 docs	expected; measure p50, not one call
Home slow *and* `home.deliveryRules > 0` in block log	Firebase	isolate cold or `config` epoch just moved	wait one call; verify epoch bump on rule save
Home slow *and* `home.weekDeliveries` large	Firebase	rider's real week of work	not amplification — do not "fix"
Admin list spins, then shows rows	Firebase	40k–200k capped scan	rollup, not smaller page
List returns fewer rows than the KPI	Firebase	capped scan hit `SCAN_CAP`	surface truncation; add rollup
Query slower after a schema change, no error	Firebase	composite index still building → bounded scan	wait for READY
Map frozen but pill says "live"	Cloudflare	rail measuring frames, not positions	45 s position-silence → `degraded` + 10 s poll
Pill says "mirror"	Cloudflare	Worker 1027 / DO cap / WS 1006	Workers Paid, or ingest over WS
Driver detail card shows km/h while Offline	Cloudflare→Firebase	stale frame printed as fact	`hasLiveTelemetry` gate

Symptom	Layer	Most likely cause	Fix
eSign compose slow, then works	Cloudflare	S3 403 → Worker fallback	provision keys; fallback is intended
`/settings/*` returns 429 in a burst	Firebase auth	rate limiter under rapid navigation	retest individually; not a user state
Anything 504 after ~25 s	Vercel	server action exceeded function budget	check sequential awaits, not payload

6. Action list, ordered by money saved per unit of risk

#	Action	Layer	Effect	Risk
1	Memoise ingest geo context per driver (60 s TTL or room `hello`)	Firebase R	removes the largest un-cached read on the write path	low — same data, shorter window
2	`limit(40)` → `orderBy('recorded_at','desc').limit(1)` in `driverReportLocation`; move `deliveryExists` after coalesce	Firebase R	40 → 1 doc per fix, and no reads on suppressed fixes	low
3	Enable Home snapshot + event-driven refresh (`record_type: 'home'`)	Firebase R	4 docs/call and single-digit calls/rider/day	medium — app release
4	Replace admin wide scans with rollups (`driver_performance_daily`, `rollups_driver_day` pattern)	Firebase R	one Order Reconciliation open currently costs more than a day of Home	medium
5	Sample `logAdminRead`; prune unreachable `deliveries` indexes	Firebase W + storage	removes a write that grows with supervision	low
6	Move to Workers Paid	Cloudflare	1027 and the "mirror" pill disappear for ~$5.60/mo	none
7	Remove `@supabase/ssr` + `supabase-js` from `package.json`	Supabase	dead dependency surface	none

Do not "fix" #6 by lowering tracking fidelity. The 1 Hz pin and the 75 m history cadence are the product; they belong to a Live Tracking decision, not to a cost ticket.

7. How to re-measure (do this before and after any change)

Question	Command / query
Fleet-wide document counts	`node --import tsx scripts/measure-firestore-amplification.mts` (read-only `count()`)
What one call actually cost	Cloud Logging: `jsonPayload.perf="handler" jsonPayload.name="<callable>"` → p50/p95 `docsRead`
Which helper is expensive	same query, `jsonPayload.perf="block"`, group by `name`
Firestore reads/writes per day	Firebase console → Firestore → Usage (compare same weekday)
The money	GCP console → Billing → Reports, service `Cloud Firestore`, group by SKU
Cloudflare requests	Workers dashboard → analytics, or GraphQL (Worker + DO **together**)
R2 operations	Cloudflare dashboard → R2 → metrics

Question	Command / query
Lane probes	`node scripts/probe-lane.mjs`

Traps that make a measurement lie: measuring a cold isolate, measuring an unauthenticated probe (`requireRider` rejects before the reads), comparing different times of day (reads track duty state), and reading `reads` instead of `docsRead`.